Zhimin Li

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Education

University of Pittsburgh, Pittsburgh, PA, USA *Jan. 2024–Present*
Ph.D. in Electrical and Computer Engineering

Northeastern University, Boston, MA, USA *Jan. 2023–Jan. 2024*
M.S. in Information Science (transfer to University of Pittsburgh)

University of Pittsburgh, Pittsburgh, PA, USA *Sept. 2019–Dec. 2022*
B.S. in Computer Science

Hefei University of Technology, Hefei, Anhui, China *Sept. 2017–Jun. 2019*
B.S. in Architectural Environment and Energy Application Engineering (transfer to University of Pittsburgh)

Research Interests

Efficient and Trustworthy AI; Edge/Embedded AI; Vision-Language Models; DNN Acceleration; AI Fairness and Privacy.

Research Experience

Privacy-Preserving Wearable Vision-Language Models *Nov. 2025–Present*
Advisor: Jingtong Hu, Pittsburgh, PA, USA

- Investigated a practical privacy risk in wearable AI (smart glasses): even when a VLM produces a harmless answer, its intermediate visual representations can still expose sensitive attributes such as location, income, sex, and personal interests.
- Designed a split-inference privacy architecture that transforms visual tokens **before they leave the trusted wearable device**, requiring joint optimization of representation disentanglement, semantic privacy suppression, and utility preservation across multiple VLM backbones.
- Built a **3,221-record privacy–utility benchmark** and reduced private-attribute inference accuracy from **56.7% to 7.4%** while retaining **74.4% utility**. Manuscript currently under review at **NeurIPS 2026**; see related publication below.

Achieve Fairness in AI-Assisted Mobile Healthcare Apps *Nov. 2023–Nov. 2025*
Advisor: Jingtong Hu, Pittsburgh, PA, USA

- Addressed demographic bias in medical AI, where models trained on imbalanced clinical datasets can perform well on average while systematically underperforming for underrepresented patient groups.
- Developed a fairness-oriented medical imaging pipeline spanning data generation, preference alignment, model optimization, and downstream diagnosis; incorporated clinician-informed preferences so that generated data remained both demographically balanced and clinically meaningful.
- Co-developed **FairGen**, improving fairness by **95.9%** on skin images, **80.0%** on chest radiography, and **35.2%** on brain MRI while maintaining competitive diagnostic accuracy. Published in **npj Digital Medicine** (IF 18); see related publication below.

Efficient Framework for Real-Time Object Detection on the Edge *Mar. 2023–Nov. 2023*
Advisor: Yanzhi Wang, Boston, MA, USA

- Tackled the gap between high-accuracy object detection models and the strict power and latency budgets of mobile devices, where processing every video frame at maximum hardware frequency wastes substantial computation and energy.
- Contributed to an adaptive deployment framework that jointly coordinates **video keyframe selection, CPU/GPU DVFS, and DNN pruning** instead of optimizing these components independently, creating a large hardware–model design space under real-time latency constraints.
- Maintained object-detection accuracy within **0.28% mAP** of processing every frame while reducing system power consumption by up to **96.7%**. Accepted by **ICCAD 2024**; see related publication below.

Digital Avatar on Mobile Device *Aug. 2023–Nov. 2023*
Advisor: Yanzhi Wang, Boston, MA, USA

- Explored how multiple generative-AI components can be integrated into a responsive digital human rather than used as isolated models, with the goal of enabling natural real-time human–AI interaction on mobile clients.
- Built an end-to-end server-to-mobile pipeline connecting **speech recognition, LLM reasoning, text-to-speech, facial animation, and audio–video streaming** through FastAPI, requiring synchronization across heterogeneous models and different inference latencies.
- Evaluated generated personalities through human preference studies; the proposed prompting strategy received **58.3% of votes** as the most humorous response. Accepted to the **IJCAI 2024 Demo Track**; see related publication below.

DNN Data Management on Hybrid Memory Systems *Feb. 2021–Sep. 2021*
Advisor: Dazhao Cheng, Charlotte, NC, USA

- Studied a systems bottleneck that becomes increasingly important as DNN models grow: fast memory offers high performance but limited capacity, while larger memory tiers introduce substantially higher data-access cost.

- Developed and evaluated data-management strategies for placing and migrating DNN data across heterogeneous memory tiers, requiring coordination between model execution, memory behavior, and runtime data movement rather than treating memory as a uniform resource.
- Contributed to system evaluation and performance analysis using ReDL; the work was accepted by **IEEE Big Data 2021**. See related publication below.

Publications

1. **Zhimin Li***, Ruichen Zhang*, Zhen Tan*, Howard J. Aizenstein, Jingtong Hu, and Tianlong Chen. "FairGen: Preference-Aligned Diffusion for Demographically Equitable Medical Image Synthesis." *npj Digital Medicine*, **IF 18.0**, 2026. doi:10.1038/s41746-026-02868-z. *Equal contribution.
2. Chao Wu, Yifan Gong, Liangkai Liu, Mengquan Li, Yushu Wu, Xuan Shen, **Zhimin Li**, Geng Yuan, Weisong Shi, and Yanzhi Wang. "AyE-Edge: Automated Deployment Space Search Empowering Accuracy yet Efficient Real-Time Object Detection on the Edge." In *Proceedings of the 43rd IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2024, Art. 178, pp. 1–9. doi:10.1145/3676536.3676655.
3. Timothy Rupprecht, Sung-En Chang, Yushu Wu, Lei Lu, Enfu Nan, Chih-hsiang Li, Caiyue Lai, **Zhimin Li**, Zhijun Hu, Yumei He, David Kaeli, and Yanzhi Wang. "Digital Avatars: Framework Development and Their Evaluation." In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24)*, Demo Track, 2024, pp. 8780–8783. doi:10.24963/ijcai.2024/1031.
4. Wei Rang, Donglin Yang, **Zhimin Li**, and Dazhao Cheng. "Scalable Data Management on Hybrid Memory System for Deep Neural Network Applications." In *2021 IEEE International Conference on Big Data (Big Data)*, 2021, pp. 1470–1480. doi:10.1109/BigData52589.2021.9671309.

Under Review

5. **Zhimin Li**, Pan Wang, Jingxian Chen, Yuantao Tang, Anthony Chen, Qian Lou, and Jingtong Hu. "Defending Wearable VLMs Against Private Attribute Inference." arXiv:2608.28691, 2026. Under review at ICLR 2027.
6. **Zhimin Li**, Pan Wang, Jingxian Chen, Yuantao Tang, Qian Lou, and Jingtong Hu. "SAGE: Efficient Protected Vision-Language Inference on Smart Glasses." arXiv:2610.14089, 2026. Under review at ICLR 2027.
7. Jingxian Chen, Shengjie Xu, **Zhimin Li**, Pan Wang, Samarjit Chakraborty, Jingtong Hu. "Safety-Aware Differentiable DNN Search for CPS Perception." Manuscript under review.
8. Pan Wang, Jie Peng, **Zhimin Li**, Jingxian Chen, Hui Ji, Huanrui Yang, Pengfei Zhou, Tianlong Chen, Jingtong Hu. "Think Before You Compute: Adaptive Budgeting and Compression for Efficient MoE-Based Multimodal Learning." Manuscript under review.

Work Experience

Northeastern University, Research Assistant, Electrical and Computer Engineering *Sep. 2023–Dec. 2023*

- Worked on DNN weight pruning, DVFS for mobile DNNs, and deployment of TTS, Stable Diffusion v1.5/v2.0, and object detection models on mobile devices.

University of Pittsburgh, Research Assistant, Electrical and Computer Engineering *May 2022–Dec. 2023*

- Parallelized CPU–FPGA DRAM data transfer with CNN inference/training and optimized the accelerator for higher throughput and lower latency.

University of Pittsburgh, Teaching Assistant, SCI–Computer Science *Fall 2022–Present*
CS 445: Data Structures; ECE 1110: Computer ORG and Architecture; ECE 1150 Computer Network.

Honors & Academic Service

Reviewer: CVPR, IJCAI, ICCAD, DAC, IEEE Transactions on Cyber-Physical Systems (TCPS).

Dean's List: 2022 Summer, 2022 Spring, 2021 Summer, 2020 Fall, 2020–2021 Spring.

Skills

Programming & Systems: Python, Linux, CUDA, Docker, Git, FastAPI.
Machine Learning & Applied AI: PyTorch, Vision-Language Models (VLMs), Multimodal Learning, Generative AI, Diffusion Models, Object Detection, Model Compression, Privacy-Preserving ML, AI Fairness.
AI Infrastructure & Edge AI: ML Systems, Model Inference, Inference Orchestration, On-Device Inference, Split Inference, GPU Computing, DNN Acceleration, DVFS, FPGA Acceleration, Hybrid Memory Systems.
Models & Frameworks: Qwen, MiniCPM, GLM, Stable Diffusion, YOLOv5, nanoGPT, Whisper.